



# Modelling Financial Markets

## Chapter 1: Stylised Facts and Returns



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## Acronyms used in this material (1/2)

- ACF: AutoCorrelation Function
- AI: Artificial Intelligence
- AR: AutoRegressive (model); in event studies: Abnormal Return
- ARCH: AutoRegressive Conditional Heteroskedasticity
- ARCH-LM: ARCH Lagrange Multiplier (test)
- BET-TR: BET Total Return (index)
- BNR: Banca Națională a României (the National Bank of Romania)
- BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- CAGR: Compound Annual Growth Rate
- CI: Confidence Interval
- CLT: Central Limit Theorem
- CRIX: CRypto IndeX
- DAX: Deutscher Aktienindex (the German stock index)
- EGARCH: Exponential GARCH
- ETF: Exchange-Traded Fund
- FTSE: Financial Times Stock Exchange (FTSE Russell, index provider)
- GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- GEO: Government Emergency Ordinance
- GJR-GARCH: Glosten-Jagannathan-Runkle GARCH
- HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- HAR-RV: Heterogeneous AutoRegressive model of Realised Volatility
- JB: Jarque-Bera (test)
- LB: Ljung-Box (test)
- MDD: Maximum DrawDown
- OHLC: Open, High, Low, Close
- PDV: Path-Dependent Volatility

## Acronyms used in this material (2/2)

- ▣ **QQ: Quantile–Quantile (plot)**
- ▣ **RMSE: Root Mean Squared Error**
- ▣ **SE: Standard Error**
- ▣ **TSPL: Time-Shifted Power Law (kernel)**
- ▣ **US: United States**
- ▣ **VIX: CBOE Volatility Index**

## Chapter Route and the Definition of a Stylised Fact

### Route of this chapter

- Part I: returns, distribution and tails, dependence and its mechanisms
- Part II
  - ▶ persistence and the limits of inference, market exceptions
  - ▶ volatility from open, high, low and close prices
  - ▶ performance, a case study
  - ▶ one pass of the AI-assisted research loop
- Technical detail sits in the [Appendix](#) at the end

- Stylised facts are the benchmark every model of returns must reproduce
  - ▶ the i.i.d. (independent and identically distributed) Normal-distribution model fails most of them

### Definition

- A **stylised fact** is a qualitative statistical regularity of **returns**
  - ▶ seen across assets, markets and periods [Cont \(2001\)](#)
- It is a hypothesis to re-test on each market and frequency [Ratliff-Crain, Van Oort, Koehler & Tivnan \(2025\)](#)
- Key message: returns show little linear autocorrelation
  - ▶ but their **size** is highly predictable

## Data Used in This Chapter

| Series  | Price used                         | Type, currency             | Calendar              | Sample          | $T$   |
|---------|------------------------------------|----------------------------|-----------------------|-----------------|-------|
| S&P 500 | daily close (plus open, high, low) | price index, USD           | US trading days       | 1990–2026       | 9 245 |
| BET-TR  | daily close                        | total return, RON          | BVB trading days      | Sep 2014 – 2026 | 2 992 |
| Bitcoin | daily close                        | price, USD                 | every day, 24/7       | Sep 2014 – 2026 | 4 384 |
| EUR/RON | BNR reference rate                 | RON per EUR, managed float | Romanian working days | Jul 2005 – 2026 | 5 352 |
| Gold    | daily close                        | spot, USD per ounce        | weekdays only         | 1990–2026       | 9 559 |

- Daily closes up to 18 September 2026; BVB: Bucharest Stock Exchange
  - ▶ daily log returns on each series' own calendar
- **Managed float:** a market-determined exchange rate with central-bank interventions
  - ▶ the official fixing of the National Bank of Romania (BNR) replaces the EUR/RON market (interbank) quotes
  - ▶ in several years these had a lag-1 autocorrelation of  $-0.4$  to  $-0.7$ , which came from quote errors
- In 1996–2005 the S&P 500 open equals the previous close on about 96% of days
  - ▶ still 16% in 2007
  - ▶ range-based estimators (built on the daily high–low range) therefore use data from 2008 on

## Returns: Definitions and Aggregation

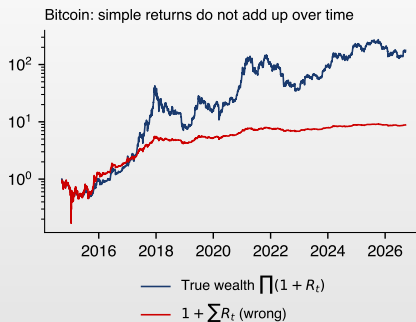
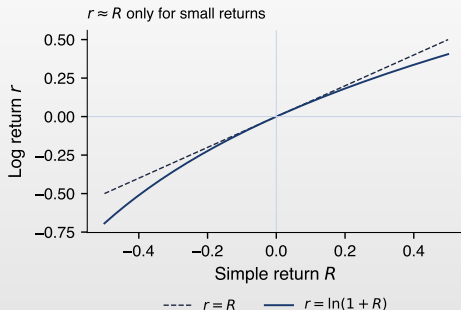
|                  | Simple return $R_t$                    | Log return $r_t$                                     |
|------------------|----------------------------------------|------------------------------------------------------|
| Definition       | $(P_t + D_t)/P_{t-1} - 1$              | $\ln(1 + R_t) = \ln P_t - \ln P_{t-1}$ (no dividend) |
| Over $k$ periods | $\prod_j (1 + R_{t-j}) - 1$            | $\sum_j r_{t-j}$ : additive over time                |
| Across assets    | $R_{p,t} = \sum_i w_i R_{i,t}$ : exact | only approximately                                   |
| Range            | $R_t \geq -100\%$                      | $(-\infty, +\infty)$                                 |
| Use              | portfolios, performance reporting      | single-asset dynamics, central limit theorem (CLT)   |

- $P_t$  is the close,  $D_t$  the dividend paid in  $(t-1, t]$ 
  - ▶ with dividends use total-return or adjusted series (BET-TR)
- The log return is always below the simple return:  $r = \ln(1 + R) < R$  for every  $R \neq 0$
- The gap is negligible for small moves and large for big falls:
  - ▶  $R = +1\%$  gives  $r = 0.995\%$
  - ▶  $R = -50\%$  gives  $r = -69.3\%$
- Textbooks: [Campbell, Lo & MacKinlay \(1997\)](#), ch. 1; [Franke, Härdle & Hafner \(2019\)](#)

## Annualisation, CAGR and Volatility Drag

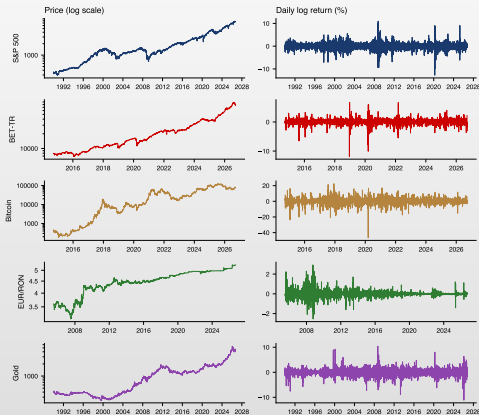
- $P$  = return intervals per year
  - ▶ 252 for stocks, 365 for crypto (traded every day)
  - ▶ gold: the actual  $\approx 260$  weekday quotes
- $\mu_{\text{ann}} = P \bar{r}$  needs only a constant mean
- $\sigma_{\text{ann}} = \sqrt{P} \hat{\sigma}$  is exact only for serially uncorrelated returns with constant variance
  - ▶ with autocorrelation the  $\sqrt{P}$  factor is no longer correct
  - ▶ EUR/RON, lag-1 autocorrelation 0.17: annual volatility rises from 4.4% to 5.3% (▶ [Appendix](#); Seminar A2)
- **CAGR** (Compound Annual Growth Rate)  $= (P_T/P_0)^{1/Y} - 1$ ,  $Y$  = number of years
  - ▶ the geometric mean  $\approx \mu_{\text{arith}} - \sigma^2/2$
  - ▶ **volatility drag**: the gap between the arithmetic and the geometric mean
  - ▶ S&P 500 1990–2026:  $\hat{\sigma} = 1.133\%$  a day, 18.0% a year; CAGR 8.69% (price index, no dividends)
  - ▶ Bitcoin 2014–2026: arithmetic mean 65.2% a year ( $P = 365$ ), CAGR 53.9%
  - ▶ a drag of 11 pp (percentage points)

## Simple and Log Returns in Practice



- Left:  $r \approx R$  only for small moves
- Right: Bitcoin 2014–2026 multiplied wealth by 176.9, i.e.  $\sum_t r_t = \ln 176.9 = 5.18$ 
  - adding daily simple returns gives  $1 + \sum_t R_t = 8.83$ , 20 times below the true multiple

## Five Markets: Prices and Returns



- Left: prices on log scale, trending and non-stationary
- Right: daily log returns fluctuate around a small mean
  - calm and turbulent periods alternate
  - extremes cluster (2008, March 2020)
- EUR/RON moves more than 1% on only 1.4% of days
- Bitcoin, 12 March 2020: log return  $-46.5\%$ 
  - a price fall of  $1 - e^{-0.465} = 37.2\%$

## Sample Moments and Their Sampling Error (1/2)

- $z_t = (r_t - \bar{r})/\hat{\sigma}$ ; skewness  $\hat{S} = \frac{1}{T} \sum_t z_t^3$ , kurtosis  $\hat{K} = \frac{1}{T} \sum_t z_t^4$ 
  - ▶ Normal distribution:  $S = 0$ ,  $K = 3$ , excess kurtosis  $K - 3 = 0$
- Mean: the standard error (SE) is  $SE(\bar{r}) = \sigma/\sqrt{T}$ 
  - ▶ daily means are 13–40 times smaller than  $\hat{\sigma}$
  - ▶ so the mean is the least precise moment
  - ▶ S&P 500:  $\bar{r} = 0.033\% \pm 0.023\%$  a day (95%),  $T = 9,245$
- i.i.d. data from the Normal distribution:  $\sqrt{T} \hat{S} \rightarrow N(0, 6)$  and  $\sqrt{T}(\hat{K} - 3) \rightarrow N(0, 24)$

## Sample Moments and Their Sampling Error (2/2)

- **Delta method:** if  $\sqrt{T}(\hat{\theta} - \theta) \rightarrow N(0, \sigma^2)$ , then  $\sqrt{T}(g(\hat{\theta}) - g(\theta)) \rightarrow N(0, g'(\theta)^2 \sigma^2)$ 
  - ▶ the standard error carries over from  $\hat{\theta}$  to  $g(\hat{\theta})$  by a first-order Taylor expansion
- Any i.i.d. distribution: the delta method gives a Normal limit
  - ▶ for  $\hat{S}$  only if  $\mathbb{E}r^6 < \infty$
  - ▶ for  $\hat{K}$  only if  $\mathbb{E}r^8 < \infty$
  - ▶ whether these moments exist is checked empirically in the section on tails

## Three Classical Tests and Their Assumptions (1/2)

- ▣ **ARCH** (AutoRegressive Conditional Heteroskedasticity): today's variance depends on past returns
  - ▶ ARCH-LM is the Lagrange multiplier test for ARCH effects
  - ▶ with  $\hat{\varepsilon}_t = r_t - \bar{r}$  and  $q = 5$
- ▣ **Volatility clustering**: days with large moves tend to follow days with large moves
- ▣ **Tail index**  $\alpha$ :  $P(|r_t| > x) \approx c x^{-\alpha}$  for large  $x$ 
  - ▶ so  $\mathbb{E}|r_t|^p < \infty$  only for  $p < \alpha$
  - ▶ the estimates in this chapter give  $\alpha \approx 3$
- ▣ We use  $m = 10$ 
  - ▶ 5% critical values:  $\chi_2^2 = 5.99$ ,  $\chi_5^2 = 11.1$ ,  $\chi_{10}^2 = 18.3$

## Three Classical Tests and Their Assumptions (2/2)

| Test                     | Statistic                                                         | $H_0$                         | $\chi^2$ limit needs             | Problem here                              |
|--------------------------|-------------------------------------------------------------------|-------------------------------|----------------------------------|-------------------------------------------|
| JB, Jarque & Bera (1980) | $T \left[ \frac{\hat{S}^2}{6} + \frac{(\hat{K}-3)^2}{24} \right]$ | i.i.d., Normal distribution   | $H_0$ itself (all moments exist) | a rejection does not say which part fails |
| LB, Ljung & Box (1978)   | $T(T+2) \sum_{k=1}^m \frac{\hat{\rho}_k^2}{T-k}$                  | $\rho_1 = \dots = \rho_m = 0$ | i.i.d., not only uncorrelated    | volatility clustering: over-rejects       |
| ARCH-LM, Engle (1982)    | $TR^2, \hat{\varepsilon}_t^2$ on $q$ lags                         | i.i.d., no ARCH effects       | $\mathbb{E}r^4 < \infty$         | $\alpha \approx 3 < 4$ : size unreliable  |

- The question is not how to run the tests
  - ▶ but whether their reference distribution holds for returns

## Worked Example: JB for the S&P 500 (1/2)

- Test statistic:  $JB = T \left( \frac{\hat{S}^2}{6} + \frac{(\hat{K} - 3)^2}{24} \right)$ 
  - ▶ distributed  $\chi^2_2$  under the null (5% critical value: 5.99)
- S&P 500, 1990–2026:  $T = 9,245$ ,  $\hat{S} = -0.364$ ,  $\hat{K} - 3 = 10.88$ 
  - ▶ skewness term  $\hat{S}^2/6 = 0.132/6 = 0.022$
  - ▶ kurtosis term  $10.88^2/24 = 118.4/24 = 4.93$
  - ▶  $JB = 9,245 \times (0.022 + 4.93) \approx 45,800$ , against the critical value 5.99
- Almost all of the statistic comes from the kurtosis term
  - ▶ the problem is the tails, not the asymmetry

## Worked Example: JB for the S&P 500 (2/2)

- Under the null all moments exist
  - ▶ so the p-value is valid for the **joint** null “i.i.d. and Normal distribution”
  - ▶ heavy tails make JB grow with  $T$
  - ▶ this is power, not a broken p-value
- A rejection does not say which part fails
  - ▶ it may be the marginal distribution or independence
  - ▶ tests of the marginal distribution with HAC (heteroskedasticity and autocorrelation consistent) variances:
    - ▶ [Appendix](#)

## Robust Alternative: Quantile-Based Skewness and Kurtosis (1/2)

- Two quantile-based measures Kim & White (2004)
  - ▶ Hinkley skewness  $\frac{q_{0.95} + q_{0.05} - 2q_{0.5}}{q_{0.95} - q_{0.05}}$
  - ▶ Crow–Siddiqui excess kurtosis  $\frac{q_{0.975} - q_{0.025}}{q_{0.75} - q_{0.25}} - 2.91$
  - ▶ both are 0 for the Normal distribution
- No moments needed
- 95% confidence interval (CI) from a **moving-block bootstrap**
  - ▶ it resamples blocks of 20 consecutive days (1,000 resamples)
  - ▶ the blocks keep the dependence within them
  - ▶ valid for stationary, weakly dependent returns with a positive density at these quantiles

## Robust Alternative: Quantile-Based Skewness and Kurtosis (2/2)

|         | Moment skew. | Hinkley [95% CI]        | Excess kurt. | Crow–Siddiqui excess [95% CI] |
|---------|--------------|-------------------------|--------------|-------------------------------|
| S&P 500 | −0.36        | −0.064 [−0.092, −0.041] | 10.9         | 1.54 [1.29, 1.79]             |
| BET-TR  | −1.40        | −0.036 [−0.085, 0.025]  | 17.8         | 1.06 [0.73, 1.43]             |
| Bitcoin | −0.70        | −0.018 [−0.069, 0.032]  | 12.0         | 2.25 [1.88, 2.64]             |
| EUR/RON | 0.73         | 0.041 [−0.022, 0.093]   | 15.1         | 4.87 [3.74, 6.13]             |
| Gold    | −0.23        | 0.004 [−0.025, 0.030]   | 9.3          | 1.47 [1.29, 1.73]             |

- ▣ Quantile (Crow–Siddiqui) excess kurtosis is positive in all five markets
  - ▶ no CI contains 0
- ▣ Quantile-based negative skewness is established only for the S&P 500
  - ▶ BET-TR's moment skewness −1.40 comes from a few days (19 December 2018)

## Summary Statistics: Five Markets

|                             | S&P 500      | BET-TR      | Bitcoin      | EUR/RON    | Gold         |
|-----------------------------|--------------|-------------|--------------|------------|--------------|
| Observations                | 9245         | 2992        | 4384         | 5352       | 9559         |
| Mean (% per day)            | 0.033        | 0.077       | 0.118        | 0.007      | 0.025        |
| Std. dev. (% per day)       | 1.13         | 0.98        | 3.49         | 0.28       | 1.01         |
| Annualised volatility (%)   | 18.0         | 15.5        | 66.6         | 4.4        | 16.3         |
| Skewness                    | -0.36        | -1.40       | -0.70        | 0.73       | -0.23        |
| Excess kurtosis             | 10.9         | 17.8        | 12.0         | 15.1       | 9.3          |
| Min / max log return (%)    | -12.8 / 11.0 | -11.9 / 6.7 | -46.5 / 22.5 | -2.5 / 2.9 | -11.1 / 10.4 |
| Jarque-Bera                 | 45805        | 40433       | 26482        | 51575      | 34732        |
| Ljung-Box $Q(10)$ , $r_t$   | 91.6         | 70.0        | 20.2         | 221.7      | 32.1         |
| Ljung-Box $Q(10)$ , $r_t^2$ | 8018         | 714         | 213          | 2420       | 1049         |
| ARCH-LM (5 lags)            | 2229         | 371         | 115          | 871        | 483          |

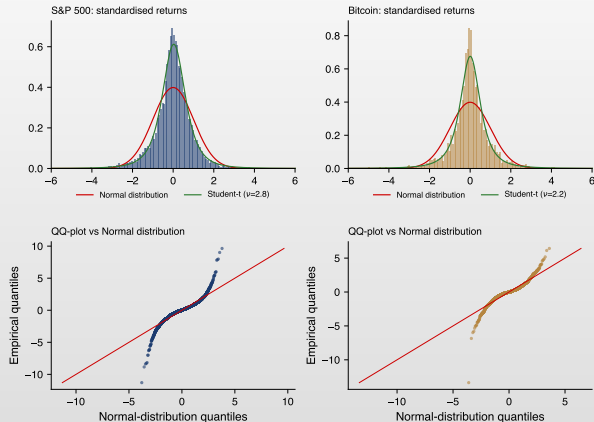
- Descriptive: heavy tails everywhere (excess kurtosis 9.3–17.8)
  - ▶  $Q(10)$  on  $r_t^2$  is 10–90 times larger than on  $r_t$
  - ▶ the size of returns is far more predictable than their direction
- Inferential: all i.i.d. p-values are below 0.03
  - ▶ but they are valid only under the assumptions in the table of the three classical tests

## The Stylised Facts of Cont (2001)

| Group                    | Facts                                                                                                                      |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Distribution             | 2. heavy tails; 3. gain/loss asymmetry; 4. aggregational Gaussianity; 7. conditional heavy tails                           |
| Dependence of returns    | 1. absence of linear autocorrelation                                                                                       |
| Dependence of volatility | 5. intermittency; 6. volatility clustering; 8. slow decay of the autocorrelation of $ r_t $ ; 11. asymmetry in time scales |
| Links to other variables | 9. leverage effect; 10. volume/volatility correlation                                                                      |

- We re-test them on five markets, with data up to September 2026
  - we add the inference each test needs
- Early evidence: Mandelbrot (1963), Fama (1965); textbook treatment: Taylor (2008), Franke, Härdle & Hafner (2019)

## Fact 2, Heavy Tails: Histograms and QQ-Plots



- Extreme returns are far more frequent than under a Normal distribution with the same variance
- Standardised returns  $(r_t - \bar{r})/\hat{\sigma}$ : higher peak and fatter tails
- QQ-plots compare empirical quantiles with those of the Normal distribution
  - ▶ they bend away from the line in both tails
- A Student- $t$  fits much better

## Tail Heaviness: Exceedances and the Student- $t$ (1/2)

- Student- $t$ , [Praetz \(1972\)](#):  $f(x) \propto \left(1 + \frac{(x-\mu)^2}{\nu s^2}\right)^{-(\nu+1)/2}$ 
  - ▶ tail  $P(X > x) \propto x^{-\nu}$ : moments exist only of order  $< \nu$
  - ▶ variance  $s^2\nu/(\nu - 2)$  for  $\nu > 2$
  - ▶ excess kurtosis  $6/(\nu - 4)$  for  $\nu > 4$
- $\hat{\nu}$  is fitted by **maximum likelihood** to the whole distribution
  - ▶ the many small returns dominate the fit
  - ▶ so a poor fit in the centre of the distribution moves  $\hat{\nu}$
  - ▶ EUR/RON:  $\hat{\nu} = 1.2$  reflects that 57% of daily changes are below 0.1%, plus rare jumps

## Tail Heaviness: Exceedances and the Student- $t$ (2/2)

|                                                                  | S&P 500 | BET-TR | Bitcoin | EUR/RON | Gold  |
|------------------------------------------------------------------|---------|--------|---------|---------|-------|
| Days beyond $3\hat{\sigma}$ (%)                                  | 1.55    | 1.44   | 1.82    | 2.06    | 1.42  |
| Days beyond $5\hat{\sigma}$ (count)                              | 30      | 12     | 10      | 29      | 26    |
| Expected under the Normal distribution, $> 5\hat{\sigma}$        | 0.005   | 0.002  | 0.003   | 0.003   | 0.005 |
| Student- $t$ degrees of freedom $\hat{\nu}$ (maximum likelihood) | 2.8     | 3.1    | 2.2     | 1.2     | 2.9   |

□ Under the Normal distribution

- ▶ 0.27% of days lie beyond  $3\sigma$  (observed 5–8 times more)
- ▶ one  $5\sigma$  day occurs every 6,900 years of trading (observed 10–30)

## The Hill Estimator of the Tail Index (1/2)

- ▣ **Tail index**  $\alpha$ :  $P(|X| > x) \approx c x^{-\alpha}$  for large  $x$ ;  $\mathbb{E}|X|^p < \infty$  only for  $p < \alpha$ 
  - ▶  $\alpha > 2$ : finite variance
  - ▶  $\alpha > 4$ : finite kurtosis
  - ▶ power laws were first argued by [Mandelbrot \(1963\)](#) and [Fama \(1965\)](#)
- ▣ **Hill estimator**, [Hill \(1975\)](#): order  $X_{(1)} \geq X_{(2)} \geq \dots \geq X_{(n)}$  and use the  $k$  largest

$$\hat{\alpha}_k = \left[ \frac{1}{k} \sum_{i=1}^k \ln X_{(i)} - \ln X_{(k+1)} \right]^{-1}, \quad \text{SE}(\hat{\alpha}_k) \approx \hat{\alpha}_k / \sqrt{k}$$

- ▣ The SE assumes independent tail observations and a Pareto tail
  - ▶ clustering of extremes makes it larger ( [Appendix](#) )

## The Hill Estimator of the Tail Index (2/2)

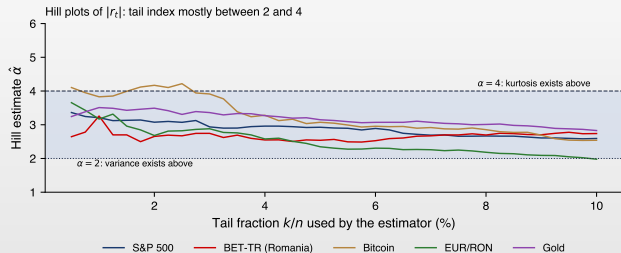
### ▣ Choice of $k$

- ▶ small  $k$  is noisy
- ▶ large  $k$  lets the centre in and biases  $\hat{\alpha}$
- ▶ read the level where the **Hill plot** ( $\hat{\alpha}_k$  against  $k$ ) is flat, here  $k = 1\text{--}5\%$  of  $n$

### ▣ Comparison with the Student- $t$ , whose tail index is $\alpha = \nu$

- ▶ EUR/RON: the Hill estimator gives  $\hat{\alpha} = 2.8$  while  $\hat{\nu} = 1.2$
- ▶ a Student- $t$  fitted to the whole distribution misdescribes the tails

## Hill Plots for Five Markets



- $\hat{\alpha}$  for  $|r_t|$  at  $k = 2.5\%$  of  $n$ 
  - S&P 500 3.1, BET-TR 2.7, Bitcoin 4.2, EUR/RON 2.8, gold 3.3
- For  $k = 1-5\%$ : between 2.3 and 3.5 except Bitcoin (3.1–4.2)
- At a common threshold, loss tails are heavier than gain tails
  - BET-TR: 2.2 versus 3.1
  - Bitcoin: 2.7 versus 3.4
- Tails are modelled with extreme value theory ([Chapter 7](#))

 MFM\_ch1\_heavy\_tails

## Tail Indices with Confidence Intervals (1/2)

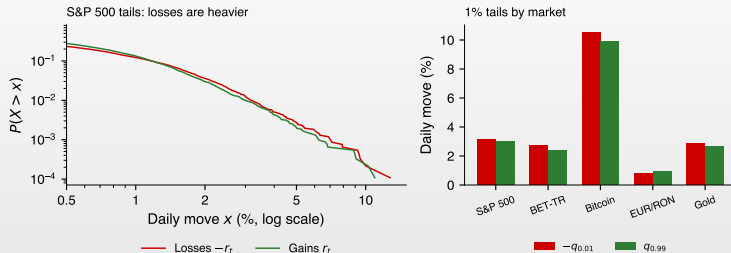
| $ r_t $ , $k = 2.5\%$ of $n$ | $n$  | $k$ | $\hat{\alpha}$ | SE i.i.d. | 95% CI (i.i.d.) | SE block bootstrap |
|------------------------------|------|-----|----------------|-----------|-----------------|--------------------|
| S&P 500                      | 9245 | 231 | 3.07           | 0.20      | [2.68, 3.47]    | 0.33               |
| BET-TR                       | 2992 | 74  | 2.67           | 0.31      | [2.06, 3.28]    | 0.45               |
| Bitcoin                      | 4384 | 109 | 4.22           | 0.40      | [3.43, 5.01]    | 0.47               |
| EUR/RON                      | 5352 | 133 | 2.82           | 0.24      | [2.34, 3.30]    | 0.28               |
| Gold                         | 9559 | 238 | 3.31           | 0.21      | [2.89, 3.73]    | 0.24               |

- Moving-block bootstrap (blocks of 20 days, 500 resamples) keeps the clustering of extremes
  - SEs are 1.1–1.6 times the i.i.d. ones

## Tail Indices with Confidence Intervals (2/2)

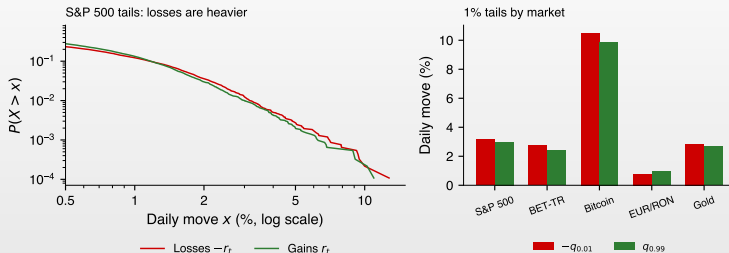
- $\alpha = 4$  is outside every interval except Bitcoin's
  - ▶ also when the interval uses the bootstrap SE
  - ▶ the fourth moment very likely does not exist
- $\alpha = 2$  is outside every i.i.d. interval
  - ▶ with the bootstrap SE only BET-TR reaches it ( $2.67 - 1.96 \times 0.45 = 1.79$ )
- Bitcoin's 4.2 uses a threshold where its Hill plot is not yet flat
  - ▶ at a common threshold its two tails give 2.71 and 3.36 (▶ [Appendix](#))

## Fact 3: Gain/Loss Asymmetry in the Data (1/2)



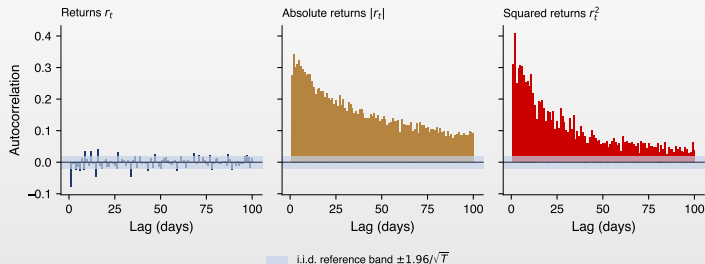
- Survival functions  $P(X > x)$  of losses  $(-r_t)$  and gains  $(r_t)$ , log-log scale
  - large losses are more frequent than gains of the same size
- VaR** (Value-at-Risk) 1% =  $-q_{0.01}$ , the loss exceeded on 1% of days (Chapter 7)

## Fact 3: Gain/Loss Asymmetry in the Data (2/2)



- VaR 1% exceeds the 99% return quantile  $q_{0.99}$  for 4 of 5 markets
  - ▶ e.g. for the S&P 500: 3.15% versus 2.97%
- EUR/RON is the exception: leu depreciations (up-moves) are larger
  - ▶  $q_{0.99} = 0.94\%$  versus  $\text{VaR } 1\% = 0.76\%$
- A Normal-distribution model with the right variance underestimates such quantiles
  - ▶ stress tests based on “ $3\sigma$  events” are too optimistic

## Facts 1 and 6: Uncorrelated but Not Independent (1/2)



□ **Fact 1:**  $\text{corr}(r_t, r_{t-k}) \approx 0$

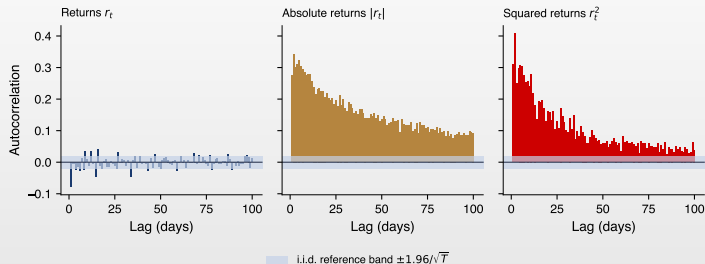
► S&P 500 1990–2026, lag 1:  $-0.08$ ; i.i.d. reference band:  $\pm 1.96/\sqrt{T}$

□ **Fact 6, volatility clustering:**  $\text{corr}(|r_t|, |r_{t-k}|) > 0$  for many lags (lag 1:  $0.27$ )

► large moves follow large moves of either sign



## Facts 1 and 6: Uncorrelated but Not Independent (2/2)



- Independence would require zero correlation of **every** function of returns
  - returns are uncorrelated but not independent
- Model  $r_t = \sigma_t \varepsilon_t$  with  $\sigma_t$  driven by past returns
  - GARCH, generalised ARCH ([Chapter 5](#))

## The ACF under Conditional Heteroskedasticity (1/2)

- ▣ **ACF** (AutoCorrelation Function):  $\hat{\rho}_k = \sum_t x_t x_{t-k} / \sum_t x_t^2$ ,  $x_t = r_t - \bar{r}$ , plotted against  $k$
- ▣ Uncorrelated returns whose conditional variance moves:  $\hat{\rho}_k \approx N(0, \tau_k)$  Romano & Thombs (1996)
  - ▶  $\hat{\tau}_k = \sum_t x_t^2 x_{t-k}^2 / (\sum_t x_t^2)^2$
  - ▶ i.i.d.:  $T\tau_k = 1$
  - ▶ volatility clustering:  $T\tau_k > 1$
- ▣ Robust portmanteau  $\tilde{Q}(m) = \sum_{k=1}^m \hat{\rho}_k^2 / \hat{\tau}_k$  Lobato, Nankervis & Savin (2001)
  - ▶ its assumptions: [▶ Appendix](#)

## The ACF under Conditional Heteroskedasticity (2/2)

| Lag 1                           | S&P 500     | BET-TR    | Bitcoin     | EUR/RON    | Gold        |
|---------------------------------|-------------|-----------|-------------|------------|-------------|
| $\hat{\rho}_1$                  | -0.079      | 0.061     | -0.022      | 0.170      | -0.011      |
| $T\hat{\rho}_1$                 | 5.00        | 4.35      | 2.89        | 6.85       | 2.43        |
| $t$ -statistic, i.i.d. / robust | -7.6 / -3.4 | 3.3 / 1.6 | -1.5 / -0.9 | 12.4 / 4.7 | -1.1 / -0.7 |

- S&P 500:  $Q(10) = 91.6$  but  $\tilde{Q}(10) = 19.0$  ( $p = 0.041$ )
- With robust SEs only the S&P 500 ( $t = -3.4$ ) and EUR/RON ( $t = 4.7$ ) keep a significant lag-1 autocorrelation
  - ▶ it is economically small

## Heavy Tails from Mixtures of Normal Distributions

- ▣ Suppose  $r_t = \sigma_t \varepsilon_t$  with  $\varepsilon_t \sim N(0, 1)$  independent of a random volatility  $\sigma_t$

$$K = \frac{\mathbb{E}[r_t^4]}{\mathbb{E}[r_t^2]^2} = 3 \frac{\mathbb{E}[\sigma_t^4]}{\mathbb{E}[\sigma_t^2]^2} \geq 3$$

- ▣ Jensen's inequality ( $\mathbb{E}[X^2] \geq \mathbb{E}[X]^2$ ) with  $X = \sigma^2$

- ▶ any variation in volatility creates excess kurtosis

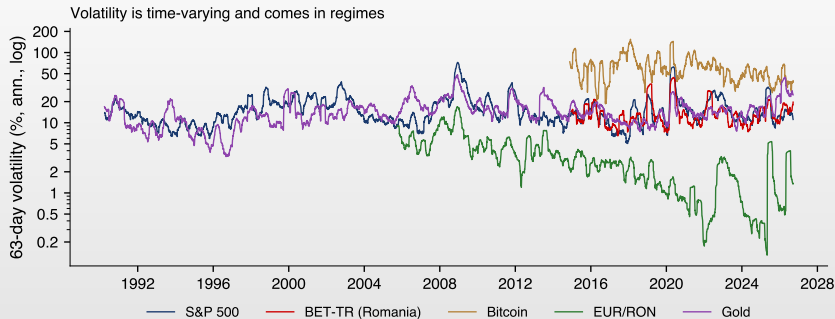
- ▣ Example:  $\sigma = 1\%$  on 90% of days and 3% on 10% of days

$$K = 3 \cdot \frac{0.9 \cdot 1 + 0.1 \cdot 81}{(0.9 + 0.1 \cdot 9)^2} = 3 \cdot \frac{9.0}{3.24} = 8.33$$

- ▣ This links heavy tails (Fact 2) to volatility clustering (Fact 6)

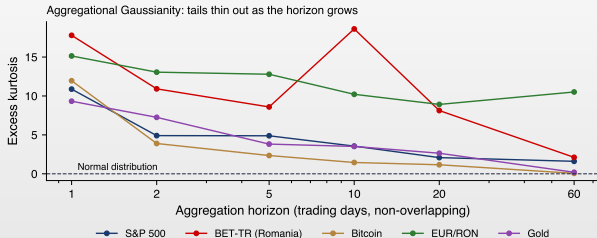
- ▶ a mixture drawn independently every day has heavy tails but no clustering (Seminar A6, A8)

## Fact 5: Intermittency in the Data



- Bursts of high volatility separated by calm periods: constant-volatility models cannot produce them
- 63-day rolling volatility  $\sqrt{P} \times$  the standard deviation over the window, log scale
  - ▶ common spikes: 2008–2009, March 2020
- EUR/RON volatility collapsed after 2012, under the BNR's managed float

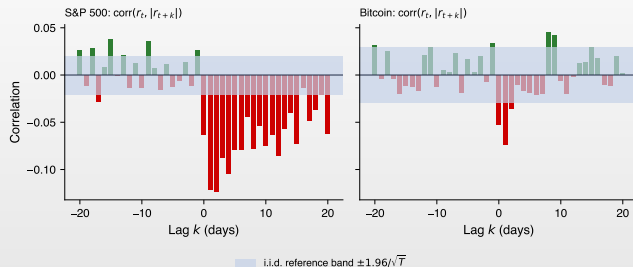
## Fact 4: Aggregational Gaussianity



- $h$ -day log return  $r_t(h) = r_t + \dots + r_{t-h+1}$  in non-overlapping blocks
  - ▶ the CLT pulls sums towards the Normal distribution
  - ▶ slowly, because daily returns are dependent
- Excess kurtosis: S&P 500  $10.9 \rightarrow 1.6$  at  $h = 60$ , Bitcoin  $12.0 \rightarrow 0.1$
- At  $h = 60$  only 154 and 73 returns remain
  - ▶ reference SE  $\sqrt{24/n} = 0.39$  and  $0.57$  under i.i.d. normality
  - ▶ so these values are very imprecise
- $h = 60$  is a quarter for the S&P 500 but two months for Bitcoin

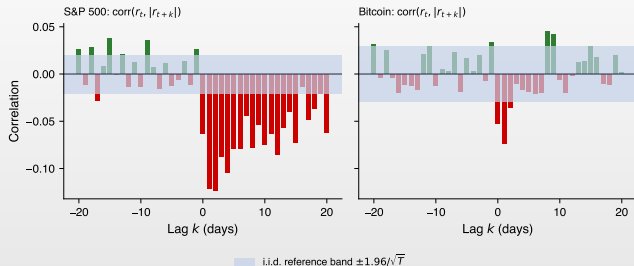
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## Fact 9: The Leverage Effect (1/2)



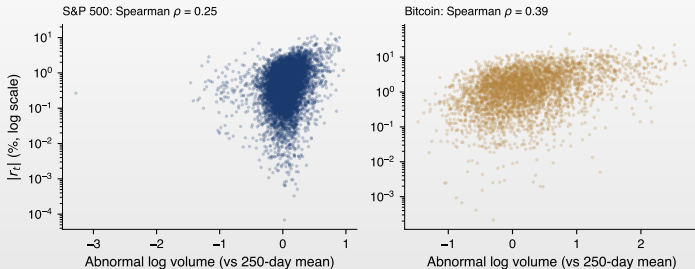
- $L(k) = \text{corr}(r_t, |r_{t+k}|)$ ,  $k > 0$ : the correlation of today's return with the size of the return  $k$  days later
- S&P 500:  $L(1) = -0.12$ , still  $-0.07$  at  $k = 10$ 
  - no effect for  $k < 0$
- BET-TR  $-0.12$ ; weak for Bitcoin ( $-0.07$ )
  - negligible for gold ( $-0.02$ ), reversed for EUR/RON ( $+0.10$ )

## Fact 9: The Leverage Effect (2/2)



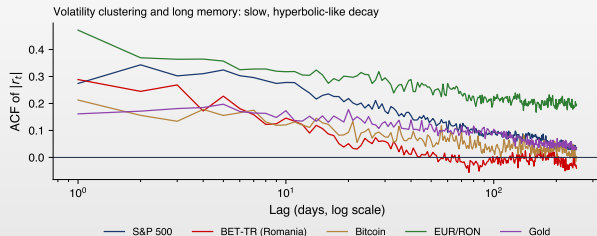
- Possible explanations
  - ▶ financial leverage of firms [Christie \(1982\)](#)
  - ▶ the retarded volatility model [Bouchaud, Matacz & Potters \(2001\)](#)
- Asymmetric GARCH models: GJR-GARCH, EGARCH ([Chapter 5](#))

## Fact 10: Volume and Volatility Move Together



- Abnormal log volume ( $\ln V_t$  minus its 250-day moving average) against  $|r_t|$ 
  - ▶ Spearman rank correlation: 0.25 (S&P 500) and 0.39 (Bitcoin)
- Mixture-of-distributions hypothesis [Clark \(1973\)](#): information flow drives both price moves and volume
  - ▶ survey of the empirical evidence: [Karpoff \(1987\)](#)
- Bitcoin volume is exchange-reported
  - ▶ it is partly **wash trading** (an exchange trading with itself) [Cong, Li, Tang & Yang \(2023\)](#)

## Fact 8: Slow Decay of the ACF of $|r_t|$ (1/2)



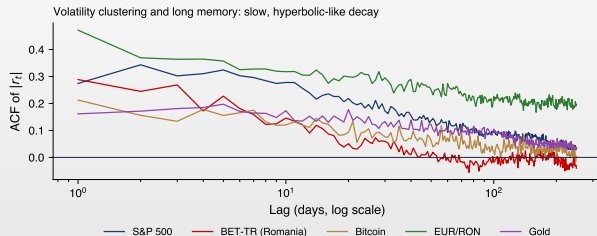
□ **Short memory:**  $\sum_k |\rho_k| < \infty$

▶ e.g. exponential decay  $\rho_k \approx c \phi^k$   
(GARCH(1,1), Chapter 5)

□ **Long memory:** hyperbolic decay  $\rho_k \approx c k^{2d-1}$ ,  
 $0 < d < \frac{1}{2}$  Ding, Granger & Engle (1993)

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## Fact 8: Slow Decay of the ACF of $|r_t|$ (2/2)



- S&P 500: 0.27 at lag 1, 0.22 at lag 20, still 0.09 at lag 100
  - ▶ BET-TR decays fastest
- Slow empirical decay is not yet established long memory
  - ▶ structural breaks, i.e. abrupt shifts in the level of volatility, can imitate it

## Estimating the Memory Parameter $d$

- Long memory: spectral density  $f(\lambda) \approx G\lambda^{-2d}$  as  $\lambda \rightarrow 0$ 
  - ▶ semiparametric estimators use only the  $m$  lowest Fourier frequencies
- Two estimators (formulas in the [Appendix](#))
  - ▶ local Whittle [Robinson \(1995\)](#):  $\sqrt{m}(\hat{d} - d) \rightarrow N(0, \frac{1}{4})$
  - ▶ log-periodogram regression [Geweke & Porter-Hudak \(1983\)](#):  $SE = \pi/\sqrt{24m}$

| $ r_t , m = T^{0.65}$          | S&P 500     | BET-TR      | Bitcoin     | EUR/RON     | Gold        |
|--------------------------------|-------------|-------------|-------------|-------------|-------------|
| $m$                            | 378         | 181         | 232         | 265         | 386         |
| Local Whittle $\hat{d}$ (SE)   | 0.48 (0.03) | 0.37 (0.04) | 0.35 (0.03) | 0.39 (0.03) | 0.40 (0.03) |
| Log-periodogram $\hat{d}$ (SE) | 0.50 (0.03) | 0.41 (0.05) | 0.35 (0.04) | 0.38 (0.04) | 0.42 (0.03) |

- $\hat{d} \approx 0.35\text{--}0.50$  everywhere
  - ▶ at least 8 SEs from  $d = 0$  (short memory) with both estimators
- The S&P 500 is close to the stationarity boundary  $\frac{1}{2}$

## Long Memory or Structural Breaks?

- Rare shifts in the level of volatility produce a slowly decaying ACF of  $|r_t|$ 
  - ▶ and  $\hat{d} > 0$  without long memory Diebold & Inoue (2001), Mikosch & Stărică (2004)
- Check: locate one variance break with the cumulative sum of squares Inclán & Tiao (1994)
  - ▶ then re-estimate  $d$  (local Whittle) on each side

| $ r_t $ | Break date  | Volatility before / after (%) | $\hat{d}$ before (SE) | $\hat{d}$ after (SE) |
|---------|-------------|-------------------------------|-----------------------|----------------------|
| S&P 500 | 20 Jul 1998 | 12.6 / 19.3                   | 0.28 (0.04)           | 0.51 (0.03)          |
| BET-TR  | 17 Dec 2018 | 12.5 / 16.9                   | 0.19 (0.05)           | 0.37 (0.04)          |
| Bitcoin | 19 Jun 2022 | 74.9 / 47.7                   | 0.39 (0.04)           | 0.15 (0.05)          |
| EUR/RON | 27 Jun 2013 | 6.6 / 2.2                     | 0.35 (0.04)           | 0.35 (0.04)          |
| Gold    | 29 Nov 2005 | 13.2 / 18.3                   | 0.30 (0.03)           | 0.37 (0.03)          |

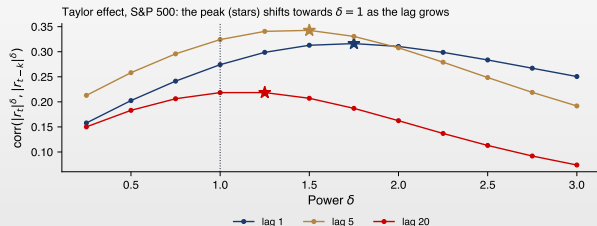
- $\hat{d}$  stays well above 0 in every sub-sample
  - ▶ one break does not explain the memory
- But  $\hat{d}$  moves (Bitcoin  $0.39 \rightarrow 0.15$ )
  - ▶ several breaks or a time-varying  $d$  remain open
  - ▶ the break dates are descriptive

## The Taylor Effect (1/2)



- **Taylor effect:** absolute returns  $|r_t|$  are more strongly autocorrelated than squared returns  $r_t^2$ 
  - ▶  $\text{corr}(|r_t|^\delta, |r_{t-k}|^\delta)$  peaks at a power  $\delta$  close to 1
  - ▶ not at  $\delta = 2$
- The chart: this autocorrelation as a function of  $\delta > 0$ 
  - ▶  $\delta = 1$ : absolute returns
  - ▶  $\delta = 2$ : squared returns

## The Taylor Effect (2/2)



- S&P 500: maximum at  $\delta = 1.75$  (lag 1), 1.5 (lag 5) and 1.25 (lag 20)
  - ▶ the optimum moves towards 1 at longer lags
  - ▶ and is never at  $\delta = 2$
- Documented by Taylor (1986), named by Granger & Ding (1995)
- The effect motivates
  - ▶ power-GARCH models, which model  $\sigma_t^\delta$  with an estimated  $\delta$  (Chapter 5)
  - ▶ absolute-value volatility measures

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## Moments, Tail Index and Valid Inference

| Statistic                                           | Needs                                                 | With Hill $\hat{\alpha} \approx 2.5\text{--}3.5$ |
|-----------------------------------------------------|-------------------------------------------------------|--------------------------------------------------|
| Mean, variance of $r_t$                             | $\alpha > 2$                                          | finite, slow convergence                         |
| ACF of $ r_t $ , robust ACF bands, $\tilde{Q}$      | $\alpha > 4$ for Normal limits                        | doubtful                                         |
| Kurtosis, ACF of $r_t^2$ , ARCH-LM, $Q$ on $r_t^2$  | $\alpha > 4$ to exist, $\alpha > 8$ for Normal limits | no standard inference                            |
| Skewness                                            | $\alpha > 6$ for a Normal limit                       | no standard inference                            |
| Quantiles, number of tail days, Hill $\hat{\alpha}$ | no moments                                            | valid, with dependence-robust SEs                |

- With an infinite fourth moment the sample ACF of  $r_t^2$  converges slowly
  - ▶ its limit is not the Normal distribution [Davis & Mikosch \(1998\)](#)
  - ▶ the  $\pm 1.96/\sqrt{T}$  band and the p-values of  $Q$  on  $r_t^2$  and ARCH-LM are not reliable
- The conclusion “volatility clusters” is robust
  - ▶ the exact magnitudes are not robust
  - ▶ one reason behind the Taylor effect:  $|r_t|$  needs fewer finite moments than  $r_t^2$

## Facts 7 and 11: Conditional Tails and Time Scales

### Fact 7: conditional heavy tails

- Fit a volatility model, e.g. GARCH (Chapter 5)
  - ▶ and standardise:  $\hat{z}_t = r_t / \hat{\sigma}_t$
- The kurtosis of  $\hat{z}_t$  is lower than that of  $r_t$ , still above 3
- Models need heavy-tailed innovations (Student- $t$ , possibly skewed)

### Fact 11: asymmetry in time scales

- Weekly volatility predicts daily volatility better than the reverse
- Traders with different horizons react at different frequencies
- The heterogeneous autoregressive model of realised volatility (**HAR-RV**, Chapter 9)
  - ▶ realised volatility is the sum of squared intraday returns

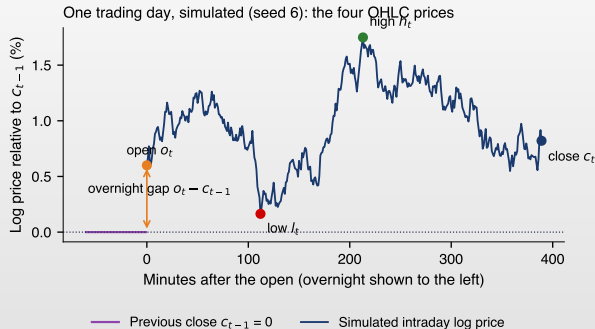
- Both facts are documented in Cont (2001)
- Returns standardised by realised volatility from intraday data are close to the Normal distribution
  - ▶ Andersen, Bollerslev, Diebold & Ebens (2001) (Chapter 9)

## Differences between the Five Markets

| Market  | Institutional feature                  | What differs                                                                                                                                                | Candidate explanation                                                                            |
|---------|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Bitcoin | trades 24/7; $P = 365$                 | volatility 66.6% versus 18.0% (S&P 500); worst day $-46.5\%$ ; weak leverage ( $-0.07$ ); ACF of $ r_t $ at lag 100: 0.02 versus 0.09                       | no corporate leverage; volatility rises after large moves of either sign                         |
| EUR/RON | BNR daily fixing, managed float        | volatility 4.4% but excess kurtosis 15.1; $\hat{\rho}_1 = +0.17$ ; largest moves October 2008 ( $+2.9\%$ , $-2.5\%$ )                                       | adjustments smoothed over several days                                                           |
| BET-TR  | total return, frontier/emerging market | skewness $-1.40$ ; worst day $-11.9\%$ on 19 Dec 2018 (bank-asset tax announced on 18 Dec, Government Emergency Ordinance 114/2018); $\hat{\rho}_1 = +0.06$ | policy shocks; non-synchronous trading of constituents<br>Campbell, Lo & MacKinlay (1997), ch. 3 |

- The explanations are hypotheses, not identified causes
- Low volatility is not low tail risk
- Heavy tails and volatility clustering hold in all five markets
  - ▶ autocorrelation and asymmetry depend on institutions

## Open, High, Low and Close (OHLC) Prices



- **OHLC:** the Open, High, Low and Close of a day
  - ▶  $o_t, h_t, l_t, c_t$ : their logs
- **Close-to-close:** the standard deviation of daily close-to-close log returns
  - ▶ one number per day
- **Range-based** estimators use the whole path
  - ▶ the range  $h_t - l_t$  is large on volatile days, even if the close equals the open
- **Efficiency:** the ratio of the variances of two estimators

## Parkinson and Garman–Klass Estimators

- ▣ **Parkinson**, [Parkinson \(1980\)](#): uses only the daily range

$$\hat{\sigma}_P^2 = \frac{1}{4 \ln 2} \frac{1}{n} \sum_{t=1}^n (h_t - l_t)^2$$

- ▣ **Garman–Klass**, [Garman & Klass \(1980\)](#): combines the range with the open-to-close move

$$\hat{\sigma}_{GK}^2 = \frac{1}{n} \sum_{t=1}^n \left[ \frac{1}{2} (h_t - l_t)^2 - (2 \ln 2 - 1) (c_t - o_t)^2 \right]$$

- ▣ Derived for a Brownian motion without drift and without overnight jumps
  - ▶ Brownian motion: the continuous-time random walk ([Chapter 11](#))
  - ▶ efficiency relative to close-to-close: about 5 (Parkinson, Seminar A5) and 7–8 (Garman–Klass)

## Rogers–Satchell and Yang–Zhang Estimators

- **Rogers–Satchell**, [Rogers & Satchell \(1991\)](#): unbiased under a non-zero **drift** (expected trend of the log price)

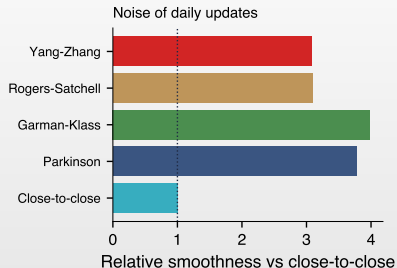
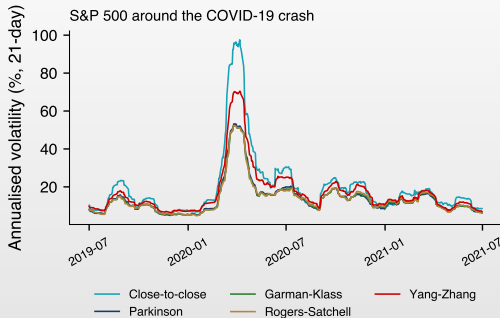
$$\hat{\sigma}_{RS}^2 = \frac{1}{n} \sum_{t=1}^n [(h_t - o_t)(h_t - c_t) + (l_t - o_t)(l_t - c_t)]$$

- **Yang–Zhang**, [Yang & Zhang \(2000\)](#): adds the overnight (close-to-open) jump

$$\hat{\sigma}_{YZ}^2 = \hat{\sigma}_{\text{overnight}}^2 + k \hat{\sigma}_{\text{open-close}}^2 + (1 - k) \hat{\sigma}_{RS}^2, \quad k = \frac{0.34}{1.34 + \frac{n+1}{n-1}}$$

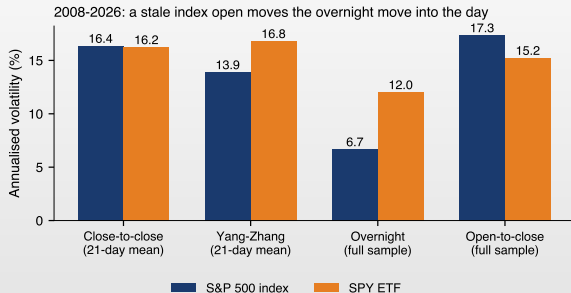
- ▶  $\hat{\sigma}_{\text{overnight}}^2$ : variance of  $o_t - c_{t-1}$
  - ▶  $\hat{\sigma}_{\text{open-close}}^2$ : variance of  $c_t - o_t$
  - ▶ minimum-variance combination among drift-independent estimators robust to opening jumps
- Range estimators are the bridge to realised volatility from intraday data ([Chapter 9](#))

## Estimators on the S&P 500



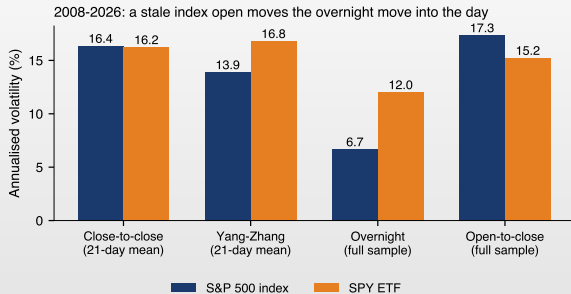
- 21-day windows, 2008–2026; averages
  - close-to-close 16.4%, Yang–Zhang 13.9%, Parkinson 13.0%, Garman–Klass 12.4%, Rogers–Satchell 12.3%
- Pure range estimators are lower: they miss overnight gaps
- The range-based series are 3–4 times smoother: a property of the estimates, not by itself proof of efficiency

## Stale Opens: the S&P 500 Index versus SPY (1/2)



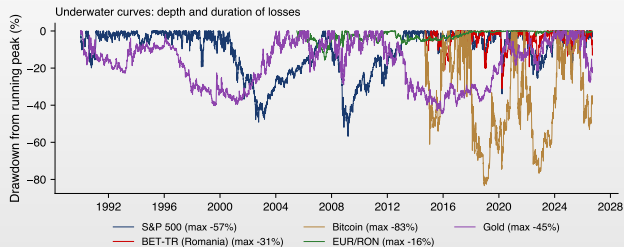
- **Stale open:** the index open is computed from the constituents' first prices
  - ▶ at 9:30 many stocks have not traded yet
  - ▶ so their previous close enters the open
- SPY, an ETF (exchange-traded fund) that tracks the S&P 500, opens at a traded price
- Open = previous close
  - ▶ about 96% of days in 1996–2005
  - ▶ 3.5% over 2008–2026 (up to 26% in 2012–2013)

## Stale Opens: the S&P 500 Index versus SPY (2/2)



- A stale open moves part of the overnight move into the day
  - ▶ index overnight volatility: 6.7%
  - ▶ SPY, whose open is a traded price: 12.0%
- Yang-Zhang falls below close-to-close on the index only
  - ▶ use a traded instrument for estimators that use the open

## Drawdowns Across Markets (1/2)



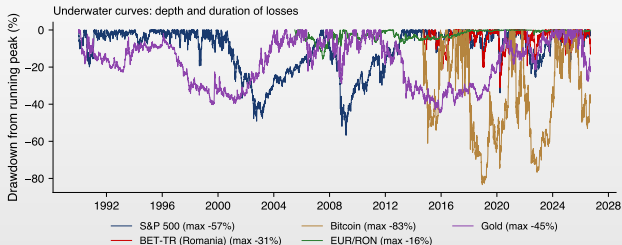
□ Drawdown  $DD_t = P_t / \max_{s \leq t} P_s - 1$  (Chapter 0)

▶ maximum drawdown:  $MDD = \min_t DD_t$

□ MDD in the five markets

▶ Bitcoin -83%, S&P 500 -57%, gold -45%  
▶ BET-TR -31% (March 2020), EUR/RON -16%

## Drawdowns Across Markets (2/2)



- S&P 500: 9 October 2007 (1,565.15) to 9 March 2009 (676.53):  $-56.8\%$ 
  - ▶ **recovery** (the price regaining its previous peak) then needs a gain of  $1/(1 - 0.568) - 1 = +131\%$
- The MDD grows with the sample length
  - ▶ compare only equal horizons ( [Appendix](#) )

## Recovery Duration after the Maximum Drawdown

- ▣ **Recovery:** the price regaining the peak it had before the fall
- ▣ **Recovery duration:** the time from that peak until the price reaches it again

|         | MDD  | Peak     | Trough   | Peak regained        |
|---------|------|----------|----------|----------------------|
| S&P 500 | −57% | Oct 2007 | Mar 2009 | Mar 2013 (5.5 years) |
| BET-TR  | −31% | Jan 2020 | Mar 2020 | Dec 2020 (0.9 years) |
| Bitcoin | −83% | Dec 2017 | Dec 2018 | Nov 2020 (3.0 years) |
| EUR/RON | −16% | Dec 2005 | Jul 2007 | Jan 2008 (2.0 years) |
| Gold    | −45% | Sep 2011 | Dec 2015 | Jul 2020 (8.9 years) |

- ▣ Depth and duration are different risks
  - ▶ gold's drawdown was shallower than Bitcoin's but lasted three times longer
- ▣ Fast V-shaped recoveries (BET-TR 2020)
  - ▶ versus long underwater periods (gold 2011–2020)
- ▣ For EUR/RON a “drawdown” is a fall of the euro against the leu

## Sharpe, Sortino and Calmar Ratios (1/2)

- **Sharpe ratio** Sharpe (1966), Sharpe (1994):  $SR = (\mu_{\text{ann}} - r_f) / \sigma_{\text{ann}}$ 
  - ▶ computed on **simple** returns, with  $\mu_{\text{ann}} = P\bar{R}$
  - ▶  $r_f = 0$  here, for comparability
- S&P 500 1990–2026:  $\mu_{\text{ann}} = 9.95\%$ ,  $\sigma_{\text{ann}} = 17.97\%$ ,  $SR = 0.55$ 
  - ▶  $\mu_{\text{ann}} = \text{mean log return } 8.33\% \text{ plus } \approx \sigma^2/2 = 1.62 \text{ pp}$

## Sharpe, Sortino and Calmar Ratios (2/2)

- ▣ **Sortino ratio** Sortino & van der Meer (1991):  $SoR = (\mu_{\text{ann}} - r_f) / \sigma_{\text{ann}}^-$ ,  $\sigma^- = \sqrt{\frac{1}{T} \sum_{t=1}^T \min(R_t, 0)^2}$ 
  - ▶ the average runs over all  $T$  days, positive days counted as 0
  - ▶ S&P 500:  $\sigma_{\text{ann}}^- = 12.68\%$ ,  $SoR = 0.78$
- ▣ **Calmar ratio** =  $\text{CAGR} / |MDD|$ 
  - ▶ S&P 500:  $8.69\% / 56.78\% = 0.15$
  - ▶ the value is driven by one episode

## Sharpe Ratios with Standard Errors (1/2)

□ Delta method, i.i.d. returns from the Normal distribution [Lo \(2002\)](#)

- ▶  $\text{Var}(\widehat{SR}_d) \approx (1 + SR_d^2/2)/T$
- ▶ annualised:  $\text{SE}(\widehat{SR}_{\text{ann}}) \approx \sqrt{P(1 + SR_d^2/2)/T} \approx 1/\sqrt{Y}$
- ▶ the span matters, not the frequency

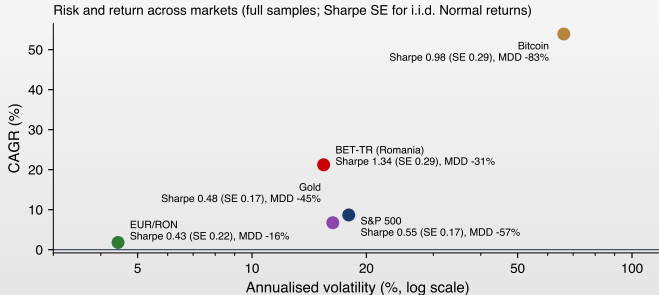
|                           | S&P 500     | BET-TR      | Bitcoin     | EUR/RON     | Gold        |
|---------------------------|-------------|-------------|-------------|-------------|-------------|
| Years $Y$                 | 36.7        | 12.0        | 12.0        | 21.2        | 36.7        |
| Sharpe ratio (SE, Normal) | 0.55 (0.17) | 1.34 (0.29) | 0.98 (0.29) | 0.43 (0.22) | 0.48 (0.17) |

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## Sharpe Ratios with Standard Errors (2/2)

- Clustering: HAC or studentised block-bootstrap tests [Ledoit & Wolf \(2008\)](#)
  - ▶ they still need finite fourth moments, which the Hill estimates put in doubt
- BET-TR versus S&P 500, common days 2014–2026
  - ▶ difference 0.61,  $z = 1.75$  ( $p = 0.08$ )
  - ▶ block-bootstrap 95% CI  $[-0.01, 1.21]$  (Seminar B13)
- Calmar has no usable sampling theory
  - ▶ its drawdown grows with the sample length

## Risk and Return Across Markets (1/2)



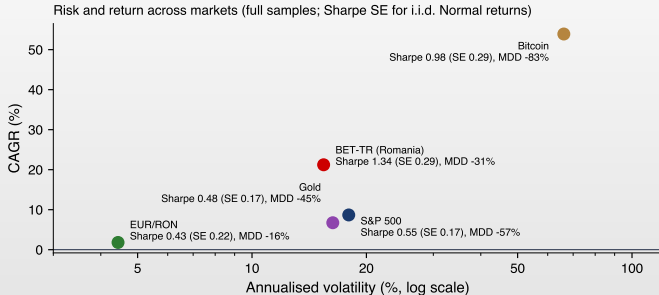
- Labels: the Sharpe ratio on simple returns and, in brackets, its standard error (SE)

►  $SE = \sqrt{P(1 + SR_d^2/2)/T}$ , valid for i.i.d. returns from the Normal distribution

► full table in the [Appendix](#)

- Different sample periods: not a ranking

## Risk and Return Across Markets (2/2)



- S&P 500 without dividends; BET-TR with dividends and in RON
  - ▶ RON short-term rates were well above USD rates
  - ▶ with  $r_f = 0$  the BET-TR Sharpe ratio is inflated
- Heavy tails and negative skewness flatter the Sharpe ratio
  - ▶ report it with drawdowns
  - ▶ and with tail-risk measures: VaR 1% and the tail index  $\hat{\alpha}$

## Case Study (1/9): Research Question and Dependent Variables

- Paper: Guyon & Lekeufack (2023), *Quantitative Finance*
  - ▶ the paper builds on Bollerslev, Hood, Huss & Pedersen (2018)
- Research question
  - ▶ what share of an equity index's volatility is explained by its past daily returns alone?
  - ▶ what functional form does this dependence take?
- Three dependent variables, modelled separately (Sections 3.4–3.5 of the paper)
  - ▶ VIX: the implied volatility of the S&P 500 over the next 30 days, computed from option prices
  - ▶ VIX9D: the same measure over a 9-day horizon
  - ▶ next-day realised volatility, computed from 5-minute returns

## Case Study (2/9): Requirement, Relevance and Data

- Requirement: a single model must reproduce simultaneously
  - ▶ volatility clustering (Fact 6)
  - ▶ the slow decay of the autocorrelation of  $|r_t|$  (Fact 8)
  - ▶ the leverage effect (Fact 9)
- Why it matters: if volatility is mostly a function of the price path, it is mostly endogenous
  - ▶ a **path-dependent volatility (PDV)** model needs little extra randomness
  - ▶ unlike stochastic volatility models ([Chapters 11–12](#))
- Data: daily closes of the S&P 500, Euro Stoxx 50, FTSE 100, DAX and Nikkei 225
  - ▶ training 1 Jan 2000 – 31 Dec 2018
  - ▶ test 1 Jan 2019 – 15 May 2022

## Case Study (3/9): Model

- Model, eq. (3.7) in the paper, with simple daily returns  $r_{t_i} = S_{t_i}/S_{t_{i-1}} - 1$

$$\sigma_t = \beta_0 + \beta_1 R_{1,t} + \beta_2 \Sigma_t, \quad R_{1,t} = \sum_{t_i \leq t} K_1(t - t_i) r_{t_i}, \quad \Sigma_t = \left( \sum_{t_i \leq t} K_2(t - t_i) r_{t_i}^2 \right)^{1/2}$$

- ▶  $\sigma_t$ : implied volatility at the close of day  $t$
- ▶  $\beta_1 < 0$  gives the leverage effect
- ▶  $\beta_2 \in (0, 1)$  gives volatility clustering, as in GARCH

## Case Study (4/9): Kernels and Estimation

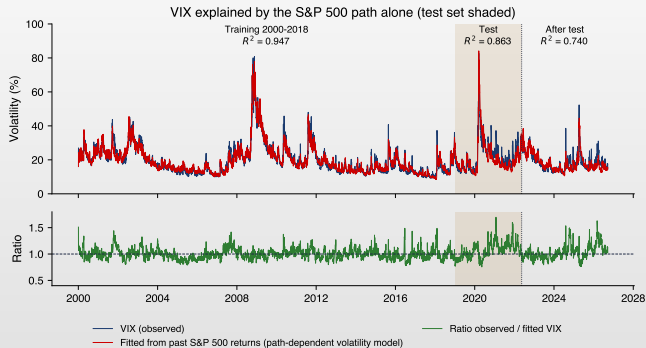
- ▣ Kernels: **time-shifted power law (TSPL)**, eq. (3.9) in the paper
  - ▶  $K(\tau) \propto (\tau + \delta)^{-\alpha}$ ,  $\alpha > 1$ ,  $\delta > 0$ , on 1,000 past days
  - ▶ large weight on the last few days (short memory)
  - ▶ a power-law tail (long memory, Fact 8)
- ▣ Seven parameters by nonlinear least squares on the training set
  - ▶ scores: RMSE (root mean squared error) and  $R^2 = 1 - \sum(\sigma_t - \hat{\sigma}_t)^2 / \sum(\sigma_t - \bar{\sigma})^2$
  - ▶ estimation details: [▶ Appendix](#)

Case Study (5/9):  $R^2$  on the Training and Test Samples

| Index                          | Implied volatility (paper, Table 2) |            | Realised volatility (paper, Table 4) |            |
|--------------------------------|-------------------------------------|------------|--------------------------------------|------------|
|                                | $R^2$ train                         | $R^2$ test | $R^2$ train                          | $R^2$ test |
| S&P 500 (VIX)                  | 0.946                               | 0.855      | 0.738                                | 0.654      |
| S&P 500 (VIX), our replication | 0.947                               | 0.863      | —                                    | —          |
| S&P 500 (9-day VIX)            | 0.876                               | 0.914      | —                                    | —          |
| Euro Stoxx 50                  | 0.929                               | 0.913      | 0.672                                | 0.682      |
| Nikkei 225                     | 0.890                               | 0.800      | 0.563                                | 0.504      |

- Past index returns alone explain 80–92% of the variance of implied volatility out of sample
  - ▶ FTSE 100 and DAX in the [Appendix](#)
- On our data (S&P 500 and VIX): RMSE 0.020 training and 0.034 test
  - ▶ versus 0.020 and 0.035 in Table 2 of the paper
- Realised volatility scores are lower
  - ▶ 5-minute realised volatility is a noisy target (Remark 3 in the paper)
  - ▶ this course uses no 5-minute data, so Table 4 of the paper is not replicated

## Case Study (6/9): Observed and Fitted VIX



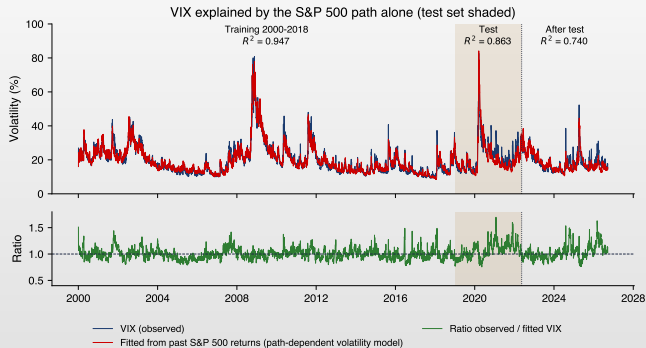
▣ Fitted on 3 Jan 2000 – 31 Dec 2018 (4,779 days)

- ▶ started from Table 3 of the paper
- ▶ then parameters frozen

▣ The test set is shaded

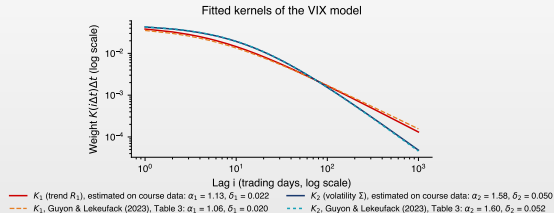
 MFM\_ch1\_path\_dependent\_vol

## Case Study (7/9): The VIX after the Test Set



- After the test set, 16 May 2022 – 18 Sep 2026:  $R^2 = 0.740$ , RMSE 0.026
  - a lower RMSE, but the VIX varied less (standard deviation 5.0 versus 9.3 points)
  - so  $R^2$  falls
- Ratio observed/fitted
  - mean 0.85 in April–May 2020
  - mean 1.16 from October 2020 to May 2022
  - 8 April 2025: VIX 52.3 versus 39.2 fitted

## Case Study (8/9): Fitted Kernels



- Daily weights of  $K_2$ , log-log scale
  - ▶  $\hat{\alpha}_2 = 1.58$ ,  $\hat{\delta}_2 = 0.050$
  - ▶ versus 1.60, 0.052 in Table 3 of the paper
- The last 5 days carry 20% of the weight
  - ▶ the days beyond lag 250 still carry 10%
  - ▶ short and long memory in one kernel

 MFM\_ch1\_path\_dependent\_vol

## Case Study (9/9): Limits and Extension

- Limits stated by the authors (Section 5 of the paper)
  - ▶ a small exogenous part remains
  - ▶ the ratio of observed to fitted VIX mean-reverts around 1
  - ▶ this ratio needs a stochastic factor
- Open extension (project seed)
  - ▶ does the same formula describe the realised volatility of Bitcoin or of BET-TR?

## From Evidence to Model Requirements

| Evidence (five markets)                                                                                         | Model requirement                                                 | Where in this course                |
|-----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|-------------------------------------|
| quantile (Crow–Siddiqui) excess kurtosis $> 0$ in all five markets; Hill $\hat{\alpha} \approx 2.5\text{--}3.5$ | heavy-tailed innovations, also conditionally                      | Ch. 5, Ch. 7 (extreme value theory) |
| ACF of $ r_t $ positive for months; $\hat{d} \approx 0.35\text{--}0.50$                                         | persistent, time-varying volatility; long memory or breaks        | Ch. 5, Ch. 9                        |
| $L(1) = -0.12$ (S&P 500, BET-TR); reversed for EUR/RON                                                          | asymmetric response to bad news where it exists                   | Ch. 5 (GJR-GARCH, EGARCH)           |
| robust $\hat{\rho}_1$ economically small except EUR/RON                                                         | approximately uncorrelated returns; market microstructure matters | Ch. 2, Ch. 10                       |
| quantile skewness negative only for the S&P 500                                                                 | asymmetry is market-specific                                      | Ch. 7–8                             |

- The random walk with i.i.d. increments from the Normal distribution (Bachelier–Black–Scholes world) violates Facts 2, 3 and 5–11
  - ▶ Fact 4 holds trivially
- ARCH-LM significance supports volatility clustering descriptively
  - ▶ with doubtful fourth moments it is not a formal proof

## Open Question: Are Tail Indices Stable over Time? (1/2)

- The tail index  $\alpha$  is usually treated as a constant
- Crypto-assets changed in structure
  - ▶ futures (2017), spot Bitcoin ETFs (January 2024), institutional holders
  - ▶ [Pele et al. \(2023\)](#): the tail factor separates cryptos from other asset classes
- Hypothesis proposed by an AI assistant
  - ▶ after the launch of the spot ETFs (January 2024), the entry of institutional investors has made the tails of Bitcoin's return distribution thinner
  - ▶ i.e. it has raised the tail index  $\alpha$

## Open Question: Are Tail Indices Stable over Time? (2/2)

- ▣ Discovery loop in one pass
  - ▶ **literature**: ask for peer-reviewed work with DOIs, then resolve each DOI yourself
  - ▶ [Hoga \(2017\)](#): change-point tests for the tail index valid under  $\beta$ -mixing dependence
  - ▶  $\beta$ -mixing: dependence that vanishes with the distance in time, e.g. in GARCH models
  - ▶ **hypotheses**: a change in tail shape, a fall in volatility only (scale does not move  $\alpha$ ), sampling noise
  - ▶ **data and code**: replicate the chapter's Hill estimates first (Bitcoin 4.2, S&P 500 3.1)
- ▣ Fixed in advance
  - ▶  $k = 2.5\%$  of  $n$ , the break date 11 January 2024
  - ▶ block-bootstrap SEs, a placebo asset

## Mini-Case: Did Bitcoin's Tail Get Thinner after the ETFs? (1/2)

- Test: Hill estimate for  $|r_t|$ ,  $k = 2.5\%$  of  $n$ 
  - ▶  $H_0 : \alpha_{\text{post}} = \alpha_{\text{pre}}$  against  $H_1 : \alpha_{\text{post}} > \alpha_{\text{pre}}$
  - ▶  $z = (\hat{\alpha}_{\text{post}} - \hat{\alpha}_{\text{pre}}) / \sqrt{s_{\text{pre}}^2 + s_{\text{post}}^2}$
  - ▶  $s$  from a moving-block bootstrap (blocks of 20 days)
- Results
  - ▶ before (Sep 2014 – 10 Jan 2024,  $k = 85$ ):  $\hat{\alpha} = 4.16$  ( $s = 0.52$ )
  - ▶ after (11 Jan 2024 – Sep 2026,  $k = 24$ ):  $\hat{\alpha} = 4.31$  ( $s = 0.91$ )
  - ▶  $z = 0.15$
- Placebo, S&P 500 at the same date:  $2.82 \rightarrow 2.86$ ,  $z = 0.03$

## Mini-Case: Did Bitcoin's Tail Get Thinner after the ETFs? (2/2)

- ▣ **Minimum detectable change**, one-sided 5% test with 80% power
  - ▶  $(1.645 + 0.842) \times 1.05 = 2.6$
  - ▶  $1.05 = \sqrt{0.52^2 + 0.91^2}$  is the SE of the difference
  - ▶ the Normal-distribution approximation is rough with 24 tail observations
  - ▶ the data cannot distinguish an unchanged tail from a large thinning of the tail
- ▣ A non-significant result does not show the absence of an effect
  - ▶ wait for more data or pool assets

## Project Seed

- **Question:** did the tail index of Bitcoin and Ethereum change with institutionalisation (futures 2017, spot ETFs 2024)?
  - ▶ raw and GARCH-standardised returns (shape versus scale)
  - ▶ loss and gain tails separately
- **Replicate first:** the chapter's Hill table and the mini-case numbers
- **Pre-register**, committed before the results
  - ▶  $H_0$ ,  $H_1$ , the rule for  $k$ , break dates and bootstrap
- **Report:** the change-point test of [Hoga \(2017\)](#) as a Quantlet
  - ▶ estimates, both SEs, the minimum detectable change
- **Log:** prompts in AI\_USE.md
  - ▶ every caught error (an i.i.d. SE, a date chosen from the data) in AI\_ERRORS.md
  - ▶ reusable prompts and checks in the [Appendix](#)

## Key Takeaways and the Link to Chapter 2

### Takeaways

- ▣ Returns are uncorrelated but not independent: their size is predictable
- ▣ Heavy tails are the rule: tail indices near 3
  - ▶ the fourth moment very likely does not exist
  - ▶ kurtosis-based inference fails
- ▣ Volatility is persistent (long memory or breaks)
  - ▶ in equity markets it reacts asymmetrically to losses
- ▣ Every test needs a reference distribution that the data may violate

### Check yourself

- ▣ Why does the  $Q(10)$  on Bitcoin returns have  $p = 0.028$  but the robust  $\tilde{Q}(10)$   $p = 0.29$ ?
- ▣ Why can the tail index be estimated with the Hill estimator when the kurtosis does not exist?
- ▣ **Chapter 2:** are returns predictable at all?

## Appendix: Formula Sheet with Assumptions

### Returns and annualisation

- ▣  $R_t = P_t/P_{t-1} - 1$ ,  $r_t = \ln(1 + R_t)$ ,  $r_t(k) = \sum_{j=0}^{k-1} r_{t-j}$ ,  
 $R_{p,t} = \sum_i w_i R_{i,t}$
- ▣  $\text{CAGR} = (P_T/P_0)^{1/Y} - 1$
- ▣  $\sigma_{\text{ann}} = \sqrt{P} \hat{\sigma}$ : uncorrelated returns, constant variance

### Tests

- ▣  $JB = T[\hat{S}^2/6 + (\hat{K} - 3)^2/24]$ :  $\chi_2^2$  under i.i.d. Normal
- ▣  $Q(m) = T(T+2) \sum_{k=1}^m \hat{\rho}_k^2 / (T-k)$ :  $\chi_m^2$  under i.i.d.
- ▣  $\tilde{Q}(m) = \sum_k \hat{\rho}_k^2 / \hat{\tau}_k$ :  $\mathbb{E}r^4 < \infty$ , zero cross terms
- ▣ Hill  $\text{SE}(\hat{\alpha}) \approx \hat{\alpha} / \sqrt{k}$ : independent Pareto tail

### Volatility

- ▣ Parkinson  $\frac{1}{4 \ln 2} (\overline{h} - l)^2$ : driftless, no overnight jumps
- ▣ Mixture  $K = 3 \mathbb{E}[\sigma^4] / \mathbb{E}[\sigma^2]^2$ :  $\sigma_t$  independent of  $\varepsilon_t$

### Performance

- ▣  $SR = (\mu - r_f) / \sigma$  on simple returns
  - $SE \approx 1/\sqrt{Y}$  for i.i.d. Normal returns
- ▣  $SoR = (\mu - r_f) / \sigma^-$
- ▣  $DD_t = P_t / \max_{s \leq t} P_s - 1$ ; Calmar =  $\text{CAGR} / |MDD|$



## Appendix: Prices Before Computers, the Stock Ticker



Stock ticker designed by Thomas Edison  
(Smithsonian collection)

Photo: Jaclyn Nash, Smithsonian National Museum of  
American History; public domain; Wikimedia Commons

- Before computers, trades reached brokers as prices printed on a paper **ticker tape**
- A daily close is one snapshot of this stream of prices
  - ▶ returns are built from such snapshots
- The choice of snapshot (close, fixing, a midnight snapshot) matters
  - ▶ as [Chapter 0](#) showed
- Prices at the level of individual trades: market microstructure, [Chapter 10](#)

## Appendix: The People Behind the Stylised Facts (1/2)



Carl Friedrich Gauss (1777–1855)

Painting: C.A. Jensen (1840); public domain;  
[Wikimedia Commons](#)

- The Normal distribution: the benchmark that returns violate



Louis Bachelier (1870–1946)

Photo: unknown author; public domain;  
[Wikimedia Commons](#)

- Prices as a random walk, in his thesis [Bachelier \(1900\)](#)

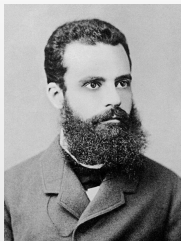


William S. Gosset, "Student" (1876–1937)

Photo: unknown author; public domain;  
[Wikimedia Commons](#)

- The Student-*t* distribution [Student \(1908\)](#), today a model for heavy tails

## Appendix: The People Behind the Stylised Facts (2/2)



Vilfredo Pareto (1848–1923)

Photo: unknown author (1870s); public domain; [Wikimedia Commons](#)



Benoît Mandelbrot (1924–2010)

Photo: Rama (2007); CC BY-SA 2.0 France; [Wikimedia Commons](#)

- Pareto: **power laws** for large incomes, the ancestor of heavy-tail models
- Mandelbrot: cotton prices have heavy tails and the Normal-distribution model fails [Mandelbrot \(1963\)](#)
- Their tail index  $\alpha$  is what the Hill estimator in this chapter measures



## Appendix: Cont's Facts in Recent Studies

- Ratliff-Crain, Van Oort, Koehler & Tivnan (2025)
  - ▶ re-test all 11 facts on intraday returns of the Dow 30 stocks, October 2018 – March 2019
  - ▶ conclusive evidence for 8 facts, no support for the remaining 3
- Heavy tails beyond the Student- $t$ : estimation of alpha-stable distributions, whose tail index is below 2 [Pele \(2014\)](#)
- Entropy-based measures of tail risk as an alternative to kurtosis [Pele, Lazar & Dufour \(2017\)](#)
- Crypto-assets: the same facts, with a tail factor that separates them from other asset classes [Pele et al. \(2023\)](#)

## Appendix: Annual Variance under Autocorrelation

- Weakly stationary returns (constant mean and variance, autocovariances depending only on the lag):

$$\text{Var}\left(\sum_{j=1}^h r_{t+j}\right) = h\sigma^2 \left[1 + 2 \sum_{j=1}^{h-1} (1 - j/h) \rho_j\right]$$

- Variance ratio**  $VR(h) = \text{Var}(h\text{-day sum})/(h\sigma^2)$ 
  - ▶ volatility over horizon  $h$  is the  $\sqrt{h}$ -rule value multiplied by  $\sqrt{VR(h)}$
- Autoregression of order 1,  $AR(1)$ , with  $\rho_j = \rho^j$ :  $\sqrt{VR(h)} \rightarrow \sqrt{(1+\rho)/(1-\rho)}$  as  $h$  grows
  - ▶ EUR/RON,  $\rho_1 = 0.17$ : factor 1.19, annual volatility 4.4%  $\rightarrow$  5.3%
  - ▶ S&P 500,  $\rho_1 = -0.079$ : factor 0.92
- Using the sample lag-1 autocorrelation as an  $AR(1)$  parameter is an assumption
  - ▶ check the whole ACF (Seminar A2)
  - ▶ variance-ratio tests: [Chapter 2](#)

## Appendix: The Null Hypothesis of JB, and Tests for Time Series

- ▣ JB is a Lagrange Multiplier test [Jarque & Bera \(1980\)](#)
  - ▶ under i.i.d. returns from the Normal distribution,  $\sqrt{T}(\hat{S}, \hat{K} - 3) \rightarrow N(0, \text{diag}(6, 24))$ , hence  $\chi^2_2$
- ▣ GARCH-type dependence changes  $\text{Var}(\hat{S})$  and  $\text{Var}(\hat{K})$ 
  - ▶ a rejection mixes the marginal distribution and independence
- ▣ [Bai & Ng \(2005\)](#): tests of skewness, kurtosis and normality of the marginal distribution of a time series
  - ▶ studentised with HAC variances
  - ▶ their separate skewness and kurtosis tests (null  $S = 0$  or  $K = 3$  without normality) need  $\mathbb{E}r^6 < \infty$  and  $\mathbb{E}r^8 < \infty$

## Appendix: The Robust Portmanteau Test and Its Assumptions

- $\tilde{Q}(m) = \sum_{k=1}^m \hat{\rho}_k^2 / \hat{\tau}_k \sim \chi_m^2$  Lobato, Nankervis & Savin (2001) needs  $\mathbb{E}r^4 < \infty$ 
  - ▶ and zero cross terms:  $\mathbb{E}[x_t^2 x_{t-k} x_{t-l}] = 0$  for  $k \neq l$
- Dalla, Giraitis & Phillips (2022) keep only the significant cross terms in the covariance of the  $\hat{\rho}_k$ 
  - ▶ none for the S&P 500, BET-TR and Bitcoin
- $\hat{\tau}_k$  is the variance under  $\rho_k = 0$ 
  - ▶ to compare a non-zero  $\rho_1$  across periods use the HAC SE of the slope of  $r_t$  on  $r_{t-1}$  (Seminar B6)
- ACF of  $|r_t|$  and of  $r_t^2$ : Normal limits need  $\alpha > 4$  and  $\alpha > 8$  Davis & Mikosch (1998)

## Appendix: Hill Estimator, Asymptotic Distribution and Alternatives

- ▣ Hill = maximum likelihood estimator of a Pareto tail above  $u = X_{(k+1)}$ , conditional on  $u$  (Seminar A11)
- ▣ If  $k \rightarrow \infty$  and  $k/n \rightarrow 0$ , and the tail is Pareto up to a second-order term

$$\sqrt{k}(\hat{\alpha}_k - \alpha) \rightarrow N(0, \alpha^2)$$

- ▶ if  $k$  grows too fast, a bias term appears: the tail is only approximately Pareto
- ▣ Dependent data (GARCH-type):  $\hat{\alpha}$  stays consistent
  - ▶ but extremes cluster and the variance exceeds  $\alpha^2/k$  [Hsing \(1991\)](#)
- ▣ Choosing  $k$ : plot  $\hat{\alpha}$  against  $\log k$  ("alt-Hill" plot) [Drees, Resnick & de Haan \(2000\)](#)
- ▣ Regression alternative [Gabaix & Ibragimov \(2011\)](#)
  - ▶ log rank on log size, with rank shifted by  $-\frac{1}{2}$
  - ▶  $SE = \hat{\alpha}\sqrt{2/k}$

## Appendix: Two Tails and the Tail of $|r_t|$

- $P(|X| > x) = P(X > x) + P(X < -x)$ 
  - ▶ with indices  $\alpha_-$  (losses) and  $\alpha_+$  (gains),  $|X|$  has index  $\min(\alpha_-, \alpha_+)$
- At the same  $k$  the estimates are not comparable
  - ▶ the  $k$  largest  $|r_t|$  come from both tails
  - ▶ common threshold  $u$ : the lower of the two one-sided thresholds at  $k = 2.5\%$  of  $n$

|         | $u$ (%) | $k$ for $ r_t $ | $\hat{\alpha},  r_t $ | $\hat{\alpha}_-, \text{losses}$ | $\hat{\alpha}_+, \text{gains}$ |
|---------|---------|-----------------|-----------------------|---------------------------------|--------------------------------|
| S&P 500 | 2.17    | 508             | 2.90                  | 2.85                            | 2.96                           |
| BET-TR  | 1.79    | 153             | 2.54                  | 2.20                            | 3.06                           |
| Bitcoin | 7.11    | 233             | 3.02                  | 2.71                            | 3.36                           |
| EUR/RON | 0.54    | 295             | 2.28                  | 2.63                            | 2.04                           |
| Gold    | 1.98    | 513             | 3.11                  | 2.97                            | 3.28                           |

- At a common  $u$ , the  $|r_t|$  estimate lies between the two tails
  - ▶ Bitcoin's 4.2 at  $k = 2.5\%$  reflects second-order bias
  - ▶ report both one-sided indices

## Appendix: Estimating $d$ , the Formulas

- ▣ Fourier frequencies  $\lambda_j = 2\pi j/T$ , periodogram  $I(\lambda_j)$  of the demeaned series
  - ▶  $m$  lowest frequencies, here  $m = T^{0.65}$
- ▣ Log-periodogram regression [Geweke & Porter-Hudak \(1983\)](#): regress  $\ln I(\lambda_j)$  on  $-2 \ln(2 \sin(\lambda_j/2))$ 
  - ▶ the slope is  $\hat{d}$ ,  $SE = \pi/\sqrt{24m}$
- ▣ Local Whittle [Robinson \(1995\)](#):

$$\hat{d} = \arg \min_d \left[ \ln \left( \frac{1}{m} \sum_{j=1}^m \lambda_j^{2d} I(\lambda_j) \right) - \frac{2d}{m} \sum_{j=1}^m \ln \lambda_j \right], \quad \sqrt{m}(\hat{d} - d) \rightarrow N(0, \frac{1}{4})$$

- ▣ Local Whittle is robust to non-Gaussianity
  - ▶ a larger  $m$  lowers the SE
  - ▶ but lets short-run (GARCH) dynamics bias  $\hat{d}$  (Seminar B15)

## Appendix: Markets in Detail (1/2)

- ▣ **Crypto-assets:** an asset class with its own statistical profile [Härdle, Harvey & Reule \(2020\)](#)
  - ▶ tail, memory and moment factors classify cryptos as a separate asset class [Pele et al. \(2023\)](#)
  - ▶ the separation comes mainly through the tail factor
  - ▶ crypto modelling and the CRIX (CRypto IndeX): [Härdle, Chen & Overbeck \(2017\)](#); Chapter 16
- ▣ **EUR/RON:** BNR reference rate from 1 July 2005
  - ▶ the new leu (RON, 1 RON = 10,000 old lei)
  - ▶ long calm periods interrupted by sudden adjustments
  - ▶ EUR/RON market (interbank) quote series showed spurious negative autocorrelation (quote errors)

## Appendix: Markets in Detail (2/2)

- ▣ **BET-TR**: the bank-asset tax was announced on 18 December 2018
  - ▶ adopted on 28 December 2018 as Government Emergency Ordinance (GEO) 114/2018
  - ▶ policy shocks produce single-day crashes without comparable rallies
- ▣ **Non-synchronous trading**: index constituents do not all trade at every moment
  - ▶ so the index reflects news with a delay
  - ▶ creating positive autocorrelation [Campbell, Lo & MacKinlay \(1997\)](#), ch. 3

## Appendix: The Maximum Drawdown of a Brownian Motion

- Driftless Brownian motion of the log price with annual volatility  $\sigma$  over  $T$  years
- The maximum **log** drawdown  $D = -\ln(1 + MDD) \geq 0$  has

$$\mathbb{E}[D] = \sqrt{\pi/2} \sigma \sqrt{T} \quad \text{Magdon-Ismail, Atiya, Pratap \& Abu-Mostafa (2004)}$$

- $MDD = e^{-D} - 1$ 
  - ▶ with a positive drift  $\mathbb{E}[D]$  grows only like  $\ln T$
- Gold 1990–2026:  $\sigma = 16.3\%$ ,  $T = 36.7$ 
  - ▶ benchmark  $\mathbb{E}[D] = 1.24$  versus observed 0.592 (−44.7%) (Seminar A4)
- Drawdowns and Calmar ratios from samples of different length are not comparable

## Appendix: Sharpe Ratios beyond the Normal Distribution

- i.i.d., any distribution with  $\mathbb{E}r^4 < \infty$  [Opdyke \(2007\)](#)
  - ▶  $\text{Var}(\widehat{SR}_d) \approx (1 - S SR_d + \frac{K-1}{4} SR_d^2) / T$
  - ▶ here the SEs change by at most 0.02
- The Deflated Sharpe Ratio corrects for non-normality and multiple testing ([Chapter 13](#))

|         | CAGR  | Sharpe | Sortino | Calmar |
|---------|-------|--------|---------|--------|
| S&P 500 | 8.7%  | 0.55   | 0.78    | 0.15   |
| BET-TR  | 21.2% | 1.34   | 1.87    | 0.68   |
| Bitcoin | 53.9% | 0.98   | 1.45    | 0.65   |
| EUR/RON | 1.8%  | 0.43   | 0.65    | 0.12   |
| Gold    | 6.7%  | 0.48   | 0.69    | 0.15   |

 MFM\_ch1\_risk\_measures

## Appendix: Case Study, Estimation Details (1/2)

- Targets (Sections 3.4–3.5 of the paper)
  - ▶ the 30-day implied volatility index of each index and the 9-day VIX
  - ▶ realised volatility of day  $t + 1$
  - ▶ volatility in decimals ( $0.03 = 3$  volatility points)
- Kernels, eq. (3.9) in the paper:  $K(\tau) = Z_{\alpha, \delta}^{-1}(\tau + \delta)^{-\alpha}$ 
  - ▶ normalised so that  $\sum_i K(i\Delta t)\Delta t = 1$  on the 1,000-day window,  $\Delta t = 1/252$
- Estimation:  $(\beta_0, \beta_1, \beta_2, \alpha_1, \delta_1, \alpha_2, \delta_2)$  by nonlinear least squares on the training set
  - ▶ initialised using a lasso regression on features built with exponential kernels

## Appendix: Case Study, Estimation Details (2/2)

- Our replication, started from the values in Table 3 of the paper (in parentheses)
  - ▶  $\hat{\beta}_0 = 0.052$  (0.057),  $\hat{\beta}_1 = -0.090$  (-0.095)
  - ▶  $\hat{\beta}_2 = 0.868$  (0.82),  $\hat{\alpha}_1 = 1.13$  (1.06)
  - ▶ the parameters differ slightly, but  $R^2$  and RMSE are practically the same
- The 9-day VIX exists only from 2011 (shorter training set)
- The restricted models ( $\beta_1 = 0$ ,  $\beta_2 = 0$ ) are Seminar C3

## Appendix: Case Study, $R^2$ for All Indices (1/2)

| Index               | Implied volatility (paper, Table 2) |            | Realised volatility (paper, Table 4) |            |
|---------------------|-------------------------------------|------------|--------------------------------------|------------|
|                     | $R^2$ train                         | $R^2$ test | $R^2$ train                          | $R^2$ test |
| S&P 500 (VIX)       | 0.946                               | 0.855      | 0.738                                | 0.654      |
| S&P 500 (9-day VIX) | 0.876                               | 0.914      | —                                    | —          |
| Euro Stoxx 50       | 0.929                               | 0.913      | 0.672                                | 0.682      |
| FTSE 100            | 0.925                               | 0.870      | 0.650                                | 0.617      |
| DAX                 | 0.934                               | 0.918      | 0.722                                | 0.557      |
| Nikkei 225          | 0.890                               | 0.800      | 0.563                                | 0.504      |

- VIX parameters:  $\beta_0 = 0.057$ ,  $\beta_1 = -0.095$ ,  $\beta_2 = 0.82$
- Across all targets  $\alpha_2 \in [1.2, 2.3]$  (Tables 3 and 5 of the paper)
  - long memory near criticality (Section 3.2 of the paper)

## Appendix: Case Study, $R^2$ for All Indices (2/2)

- Dropping  $\Sigma_t$  hurts the most
- The model beats two ARCH benchmarks and four published path-dependent models
  - ▶ Figures 3.8–3.9 and Appendix C of the paper
- Its Markovian 4-factor version, driven by Brownian motion (Section 4.6 and Table 9 of the paper)
  - ▶ gives estimated values of about 0.12–0.17 of the Hurst exponent  $H$  for daily realised volatility, as in data
  - ▶  $H < 0.5$ : a path “rougher” than Brownian motion
  - ▶ a model without rough volatility reproduces the small values of  $H$  estimated from daily data

## Appendix: The Discovery Loop, Reusable Prompts (1/2)

### □ Literature

- ▶ Find peer-reviewed work since 2015 on time variation or structural breaks in tail indices of asset returns, with tests valid under GARCH-type dependence.
- ▶ For each: DOI, estimator, test, data.
- ▶ Flag entries you are unsure of.

### □ Hypothesis

- ▶ State competing hypotheses for a change in Bitcoin's tail after the 2024 spot ETFs.
- ▶ What does each imply for the Hill estimate of the tail index of raw and of GARCH-standardised returns?

## Appendix: The Discovery Loop, Reusable Prompts (2/2)

### □ Data and code

- ▶ Use daily Bitcoin closing prices in USD; log returns on its own calendar;
- ▶ Hill estimate of the tail index for  $|r|$ , with  $k = 2.5\%$  of  $n$ , before and after 11 January 2024; moving-block bootstrap standard errors.

### □ Robustness

- ▶ Write a pre-registered plan:  $k$  from 1% to 5% of  $n$ , loss and gain tails, GARCH-standardised residuals, block lengths 10 to 50 days, a placebo date and a placebo asset.

### □ Critique

- ▶ Referee this design.
- ▶ With 24 tail observations after the break, what is the power against a change of 1 in the tail index?

## Appendix: Mandatory Checks for the Researcher (1/2)

- **Wrong return or price series:** reproduce the chapter's Hill table before anything else
- **Choice of  $k$  after seeing the result**
  - ▶ fix the rule ( $k = 2.5\%$  of  $n$ ) in advance
  - ▶ show the Hill plot for 1–5%
- **i.i.d. standard errors:**  $\hat{\alpha}/\sqrt{k}$  ignores volatility clustering
  - ▶ compare with a block bootstrap
  - ▶ for the S&P 500 before 2024 the bootstrap SE is 2.2 times larger

## Appendix: Mandatory Checks for the Researcher (2/2)

- **Break date chosen from the data**
  - ▶ use a date fixed by the event
  - ▶ or a sup-type change-point test with its own critical values [Hoga \(2017\)](#)
- **Bad ticks create fake tails**
  - ▶ list the ten largest  $|r_t|$  and match each to a dated event
- **Low power:** report the minimum detectable change
  - ▶ 2.8 SEs of the difference for a two-sided 5% test with 80% power
  - ▶ 2.5 SEs for a one-sided test

## Appendix: Practical Checklist for Empirical Work (1/2)

- ▣ **Know your data**
  - ▶ price vs total-return index
  - ▶ adjusted vs raw prices
  - ▶ fixing vs traded price
  - ▶ official reference rates where possible (EUR/RON)
- ▣ **Clean before you model**
  - ▶ non-trading days, holidays, stale quotes, obvious errors
  - ▶ match the largest moves to real events
- ▣ **Use log returns** for single-asset dynamics, simple returns for portfolios

## Appendix: Practical Checklist for Empirical Work (2/2)

- ▣ **Report robust statistics:** quantiles, the number of tail days and the tail index  $\hat{\alpha}$  next to kurtosis
- ▣ **Annualise carefully**
  - ▶ 252 vs 365 periods
  - ▶ the  $\sqrt{P}$  rule needs uncorrelated returns with constant variance
- ▣ **Reproducibility:** code and data on GitHub, as Quantlets

## Appendix: Further Study on Quantinar

### Advanced courses on quantinar.com

- ▣  Statistics of Financial Markets
- ▣  Tukey's g- and h- transformations
- ▣  Applied Time Series Analysis with Python

### Code, data and essential reading

- ▣ All charts and tables: [github.com/danpele/MFM/tree/main/Quantlets/Ch\\_01](https://github.com/danpele/MFM/tree/main/Quantlets/Ch_01)
- ▣ Essential reading: Cont (2001); Franke, Härdle & Hafner (2019); Campbell, Lo & MacKinlay (1997), ch. 1–2

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